Lecture 1:
Propositional Logic

Propositions

P, Q, A, X…

L : The King of Spain is a Leprechaun

W: Water is Wet

B : The King of Spain is a Bear

Negation (Not)

P, ¬P

¬ W = Water is not Wet

Conjunction (AND)

$P \land Q$

$L \land W$

True if the King of Spain is a Leprechaun and Water is Wet

False Otherwise

Disjunction (OR)

$P \vee Q$

“My way or the highway”

$W \vee L$

True if water is wet
True if KoS is Lep..
False if neither

Implication (IF-THEN)

$P \rightarrow Q$

If P is true then Q is true

$W \rightarrow L$

True if water isn't wet
True if the King of Spain is a leprechaun
False if W but not L

Exclusive-Or (XOR)

$P \veebar Q := (P \land \neg Q) \lor (\neg P \land Q)$

Either P is true or Q is but not both.

If and only iff (IFF)

$P \leftrightarrow Q := (P \rightarrow Q) \land (Q \rightarrow P)$

True iff (P and Q are both true) or (P and Q are both false).